WEEK-4

AIM:

Implementation of the following Memory Allocation Methods for fixed partition

a)First Fit b) Worst Fit c) Best Fit

ALGORITHM:

Step 1: Start

Step 2: Input

- Read the number of memory partitions n.
- Read the size of each partition into the original[] array.
- Read the number of processes m.
- Read the size of each process into the processes[] array.

Step 3: First Fit

1. Copy the original partition sizes into the first[] array.
2. Initialize all process allocations to -1.
3. For each process, search the partitions from the beginning.
4. If a partition is free and its size is greater than or equal to the process size:
   - Allocate the process to that partition.
   - Store the partition number.
   - Mark the partition as occupied by setting its size to -1.
   - Stop searching for that process.
5. If no suitable partition is found, mark the process as *Not Allocated*.
6. Display the First Fit allocation.

Step 4: Best Fit

1. Copy the original partition sizes into the best[] array.
2. Initialize all process allocations to -1.
3. For each process, search all free partitions.

4. Find the *smallest partition* whose size is greater than or equal to the process size.

5. If a suitable partition is found:

- Allocate the process to that partition.
- Store the partition number.
- Mark the partition as occupied by setting its size to -1.

6. If no suitable partition is found, mark the process as *Not Allocated*.

7. Display the Best Fit allocation.

Step 5: Worst Fit

1. Copy the original partition sizes into the worst[] array.

2. Initialize all process allocations to -1.

3. For each process, search all free partitions.

4. Find the *largest partition* whose size is greater than or equal to the process size.

5. If a suitable partition is found:

* Allocate the process to that partition.

* Store the partition number.

* Mark the partition as occupied by setting its size to -1.

6. If no suitable partition is found, mark the process as *Not Allocated*.

7. Display the Worst Fit allocation.

Step 6: Stop

PROGRAM

```
#include <stdio.h>
void firstFit(int partitions[], int n, int processes[], int m)
{
   int allocation[m];

   for (int i = 0; i < m; i++)
      allocation[i] = -1;

   for (int i = 0; i < m; i++)
   {
      for (int j = 0; j < n; j++)
      {
         if (partitions[j] >= processes[i])
         {
            allocation[i] = j;
            partitions[j] = -1;  // Partition is occupied
            break;
         }
      }
   }

   printf("\n--- First Fit ---\n");
   printf("Process\tSize\tPartition\n");

   for (int i = 0; i < m; i++)
   {
      if (allocation[i] != -1)
         printf("P%d\t%d\t%d\n", i + 1, processes[i], allocation[i] + 1);
```

```
        else
            printf ("P%d\t%d\tNot Allocated\n", i + 1, processes[i]);
    }
}

void bestFit(int partitions[], int n, int processes[], int m)
{
    int allocation[m];
        for (int i = 0; i < m; i++)
        allocation[i] = -1;

    for (int i = 0; i < m; i++)
    {
        int best = -1;

        for (int j = 0; j < n; j++)
        {
            if (partitions[j] >= processes[i])
            {
                if (best == -1 || partitions[j] < partitions[best])
                    best = j;
            }
        }

        if (best != -1)
        {
            allocation[i] = best;
            partitions[best] = -1;
        }
```

```
    }

    printf("\n--- Best Fit ---\n");
    printf("Process\tSize\tPartition\n");

    for (int i = 0; i < m; i++)
    {
        if (allocation[i] != -1)
            printf("P%d\t%d\t%d\n", i + 1, processes[i], allocation[i] + 1);
        else
            printf("P%d\t%d\tNot Allocated\n", i + 1, processes[i]);
    }
}

void worstFit(int partitions[], int n, int processes[], int m)
{
    int allocation[m];

    for (int i = 0; i < m; i++)
        allocation[i] = -1;

    for (int i = 0; i < m; i++)
    {
        int worst = -1;

        for (int j = 0; j < n; j++)
        {
            if (partitions[j] >= processes[i])
            {
```

```
            if (worst == -1 || partitions[j] > partitions[worst])
                worst = j;
        }
    }

    if (worst != -1)
    {
        allocation[i] = worst;
        partitions[worst] = -1;
    }
}

printf("\n--- Worst Fit ---\n");
printf("Process\tSize\tPartition\n");

for (int i = 0; i < m; i++)
{
    if (allocation[i] != -1)
        printf("P%d\t%d\t%d\n", i + 1, processes[i], allocation[i] + 1);
    else
        printf("P%d\t%d\tNot Allocated\n", i + 1, processes[i]);
}
}

int main()
{
    int n, m;

    printf("Enter number of partitions: ");
    scanf("%d", &n);
```

```
int original[n];

  printf("Enter size of each partition:\n");
  for (int i = 0; i < n; i++)
  {
     printf("Partition %d: ", i + 1);
     scanf("%d", &original[i]);
  }

  printf("\nEnter number of processes: ");
  scanf("%d", &m);

  int processes[m];

  printf("Enter size of each process:\n");
  for (int i = 0; i < m; i++)
  {
     printf("Process %d: ", i + 1);
     scanf("%d", &processes[i]);
  }

  // First Fit
  int first[n];
  for (int i = 0; i < n; i++)
     first[i] = original[i];

  firstFit(first, n, processes, m);
```

```
    // Best Fit
    int best[n];
    for (int i = 0; i < n; i++)
        best[i] = original[i];

    bestFit(best, n, processes, m);

    // Worst Fit
    int worst[n];
    for (int i = 0; i < n; i++)
        worst[i] = original[i];

    worstFit(worst, n, processes, m);

    return 0;
}
```

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Name of the Laboratory:

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

OUTPUT:

TESTCASE 1:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\hars
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc FITS.C -o FITS } ;
f ($?) { .\FITS }
Enter number of partitions:

                          5
Enter size of each partition:
Partition 1: 25
Partition 2: 69
Partition 3: 6
Partition 4: 78
Partition 5: 96

Enter number of processes: 3
Enter size of each process:
Process 1: 69
Process 2: 98
Process 3: 97

--- First Fit ---
Process Size    Partition
P1      69      2
P2      98      Not Allocated
P3      97      Not Allocated

--- Best Fit ---
Process Size    Partition
P1      69      2
P2      98      Not Allocated
P3      97      Not Allocated

--- Worst Fit ---
Process Size    Partition
P1      69      5
P2      98      Not Allocated
P3      97      Not Allocated
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

:

TESTCASE 2:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc FITS.C -o FITS } ;
f ($?) { .\FITS }
Enter number of partitions: 5
Enter size of each partition:
Partition 1: 69
Partition 2: 98
Partition 3: 97
Partition 4: 96
Partition 5: 99

Enter number of processes: 5
Enter size of each process:
Process 1: 88
Process 2: 55
Process 3: 66
Process 4: 33
Process 5: 22

--- First Fit ---
Process Size    Partition
P1      88      2
P2      55      1
P3      66      3
P4      33      4
P5      22      5

--- Best Fit ---
Process Size    Partition
P1      88      4
P2      55      1
P3      66      3
P4      33      2
P5      22      5

--- Worst Fit ---
Process Size    Partition
P1      88      5
P2      55      2
P3      66      3
P4      33      4
P5      22      1
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE 3:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc FITS.C -o FITS } ;
f ($?) { .\FITS }
Enter number of partitions: 4
Enter size of each partition:
Partition 1: 25
Partition 2: 69
Partition 3: 68
Partition 4: 59

Enter number of processes: 5
Enter size of each process:
Process 1: 56
Process 2: 36
Process 3: 23
Process 4: 31
Process 5: 21

--- First Fit ---
Process Size    Partition
P1      56      2
P2      36      3
P3      23      1
P4      31      4
P5      21      Not Allocated

--- Best Fit ---
Process Size    Partition
P1      56      4
P2      36      3
P3      23      1
P4      31      2
P5      21      Not Allocated

--- Worst Fit ---
Process Size    Partition
P1      56      2
P2      36      3
P3      23      4
P4      31      Not Allocated
P5      21      1
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```